Diploma in Electronics / Electronics & Communication Engineering • Semester 3

Principles of Electronic Communication Lab

Hands-on laboratory course for communication components, AM/FM generator and demodulator circuits, pulse modulation circuits and FM receiver-related stages.

Course Code: 3046 | **Category: Program Core** | **Credits: 1.5** | **L:T:P = 0:0:3** | **Lab / Practical**

COURSE OVERVIEW

Principles of Electronic Communication Lab (3046)

This is a standalone digital textbook, revision guide, workbook and answer guide prepared from the uploaded official syllabus PDF.

Student-friendly summary

Hands-on laboratory course for communication components, AM/FM generator and demodulator circuits, pulse modulation circuits and FM receiver-related stages.

Study tip: □□□ module-□□ separate □□□ □□□□□□□□□. Definitions, diagrams, formulas/procedures, and expected answers English technical terms □□□□□□□□□□ practise □□□□□□□□.

Course Metadata

ITEM	DETAILS
Course Code	3046
Base Course Code	3046

ITEM	DETAILS
Course Title	Principles of Electronic Communication Lab
Programme / Department	Diploma in Electronics / Electronics & Communication Engineering
Semester	3
Course Category	Program Core
Credits	1.5
L:T:P	0:0:3
Periods / semester	45
Subject Type	Lab / Practical

Course Objectives

- Give hands-on experience with electronic components, analogue modulation circuits and pulse modulation circuits.
- Demonstrate the working of various stages of FM and solve associated problems.
- Guide students in developing an FM receiver of high fidelity.

Prerequisite Knowledge

- Testing active/passive components and equipment familiarization - Fundamentals of Electrical and Electronics Engineering Lab
- Operation of transistor and diode - Basic Electronics

Course Outcomes

CO	DESCRIPTION	HOURS	LEVEL
CO1	Illustrate working of electronic components in communication system.	3	Understanding
CO2	Experiment with analog modulation schemes.	12	Applying
CO3	Develop pulse modulation circuits.	9	Applying
CO4	Build different stages in AM and FM.	15	Applying

Module Map

1. **Module 1:** Communication components familiarization (3 h)
2. **Module 2:** Analog modulation experiments (12 h)

3. **Module 3:** Pulse modulation circuits (9 h)
4. **Module 4:** AM/FM stages and open-ended FM receiver (15 h)

How to study this subject

- Read the aim and apparatus before lab.
- Draw the circuit/setup diagram before wiring.
- Write procedure and observation table in advance.
- After experiment, write result and precautions immediately.
- Practise viva questions module-wise.

Exam preparation method

- Memorise lab record format: aim, apparatus, theory, diagram, procedure, observation, result.
- Practise neat wiring/setup explanation.
- Prepare safety precautions and viva answers.
- Know how to calculate/read final result from observations.

Quick Navigation Cards

EXPERIMENT BANK

Experiment Bank - Principles of Electronic Communication Lab

Quick reference cards for revision. Use this only after reading the module explanation.

Record format

Aim -> Apparatus -> Circuit -> Procedure -> Observation -> Result

Meaning: Standard practical record structure.

Where used: All lab experiments.

Common mistake: Leaving observation table incomplete.

AM index from envelope

m=(Vmax-Vmin)/(Vmax+Vmin)

Meaning: Find modulation index from CRO waveform.

Where used: AM generator experiment.

Common mistake: Measuring Vmax/Vmin from wrong points.

555 IC role

Timer / pulse generator

Meaning: Used for FM/PWM/PPM circuits.

Where used: Pulse modulation experiments.

Common mistake: Wrong power pins or reset pin.

565 IC role

PLL IC

Meaning: Used for FM demodulation/frequency multiplier.

Where used: FM generator/demodulator and multiplier.

Common mistake: Not setting lock range correctly.

FM receiver band

80 MHz - 108 MHz

Meaning: Open-ended FM receiver setup range.

Where used: Open-ended experiment.

Common mistake: Wrong antenna/tuning range.

MODULE 1 • 3 HOURS

Communication components familiarization

Familiarize 555, 565, LM380 and counter IC 7490 used in communication circuits.

Learning outcome

Familiarize 555, 565, LM380 and counter IC 7490 used in communication circuits.

Malayalam support: □□□□□ meaning □□□□□□□□□□□□□, □□□□□□ English technical terms □□□□□□□□□□ answer □□□□□□.

Important terms

- 555 timer IC
- 565 PLL IC
- LM380 audio amplifier
- Counter IC 7490
- Pin identification and datasheet reading

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M1.01	Familiarize 555, 565, LM380 and Counter IC 7490	3	Understanding

Topic-wise explanation

555 timer IC: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

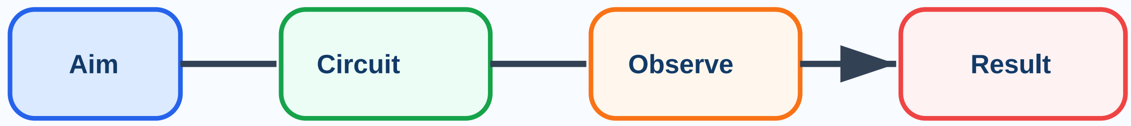
565 PLL IC: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

LM380 audio amplifier: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Counter IC 7490: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Pin identification and datasheet reading: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Visual block / diagram idea



Communication components familiarization: lab record and practical workflo

Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

Practical example: For AM generation, set carrier and modulating signals, observe output envelope, measure V_{max} and V_{min}, and calculate modulation index $m = (V_{max} - V_{min})/(V_{max} + V_{min})$.

Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

Expected Questions

M1-Q1 Write the aim and apparatus required for an experiment based on Communication components familiarization.

M1-Q2 Explain the step-by-step procedure for Communication components familiarization.

M1-Q3 Draw or describe the required circuit/setup and observation table for Communication components familiarization.

M1-Q4 List important precautions and safety points while doing Communication components familiarization.

M1-Q5 Write viva questions with answers for Communication components familiarization.

M1-Q6 How will you write the final result and verify correctness in Communication components familiarization?

Module checklist

- ✓ Aim and apparatus are clear.
- ✓ Circuit/setup is neat and labelled.
- ✓ Procedure is in correct order.
- ✓ Observation table and result are complete.
- ✓ Precautions and viva points are prepared.

MODULE 2 • 12 HOURS

Analog modulation experiments

Construct and observe AM/FM generator and demodulator circuits.

Learning outcome

Construct and observe AM/FM generator and demodulator circuits.

Malayalam support: □□□□□ meaning □□□□□□□□□□□□□, □□□□□□ English technical terms □□□□□□□□□□ answer □□□□□□.

Important terms

- AM generator using transistor
- Modulation index measurement
- AM demodulator using diode
- Input-output waveform observation
- FM generator using 555 IC
- FM generator and demodulator using 565 IC

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M2.01	Construct AM generator using transistor and measure modulation index	3	Applying

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M2.02	Construct AM demodulator using diode and observe waveforms	3	Applying
M2.03	Build FM generator using 555 IC and observe waveforms	3	Applying
M2.04	Build FM generator and demodulator using 565 IC	3	Applying

Topic-wise explanation

AM generator using transistor: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Modulation index measurement: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

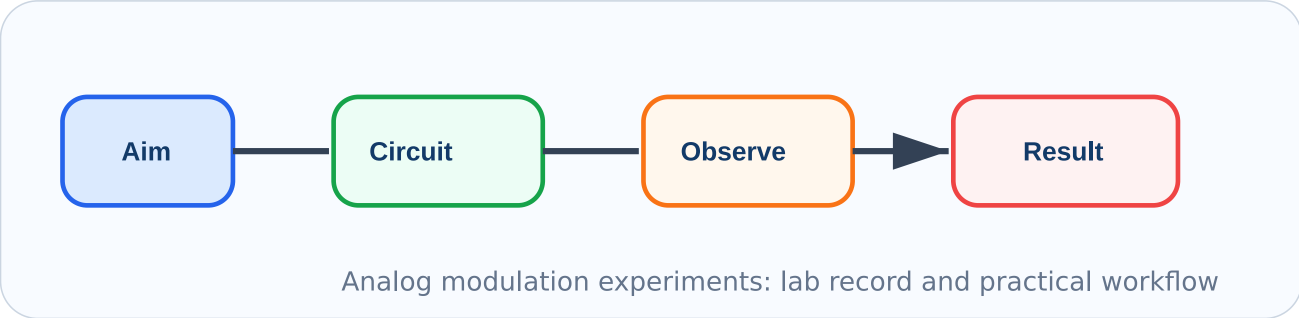
AM demodulator using diode: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Input-output waveform observation: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

FM generator using 555 IC: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

FM generator and demodulator using 565 IC: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Visual block / diagram idea



Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

Practical example: For AM generation, set carrier and modulating signals, observe output envelope, measure V_{max} and V_{min}, and calculate modulation index $m = (V_{max} - V_{min})/(V_{max} + V_{min})$.

Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

Expected Questions

M2-Q1 Write the aim and apparatus required for an experiment based on Analog modulation experiments.

M2-Q2 Explain the step-by-step procedure for Analog modulation experiments.

M2-Q3 Draw or describe the required circuit/setup and observation table for Analog modulation experiments.

M2-Q4 List important precautions and safety points while doing Analog modulation experiments.

M2-Q5 Write viva questions with answers for Analog modulation experiments.

M2-Q6 How will you write the final result and verify correctness in Analog modulation experiments?

Module checklist

✓ Aim and apparatus are clear.

✓ Circuit/setup is neat and labelled.

✓ Procedure is in correct order.

✓ Observation table and result are complete.

✓ Precautions and viva points are prepared.

MODULE 3 • 9 HOURS

Pulse modulation circuits

Develop PAM, PWM and PPM modulator/demodulator circuits and observe waveforms.

Learning outcome

Develop PAM, PWM and PPM modulator/demodulator circuits and observe waveforms.

Malayalam support: □□□□□ meaning □□□□□□□□□□□□□, □□□□□□ English technical terms □□□□□□□□□□ answer □□□□□□.

Important terms

- PAM modulator and demodulator
- PWM modulator using 555 IC
- PPM modulator using 555
- Input-output waveform comparison
- Duty cycle and pulse position observation

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M3.01	Develop PAM modulator and demodulator circuit	3	Applying
M3.02	Develop PWM modulator using 555 IC	3	Applying

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M3.03	Develop PPM modulator using 555	3	Applying

Topic-wise explanation

PAM modulator and demodulator: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

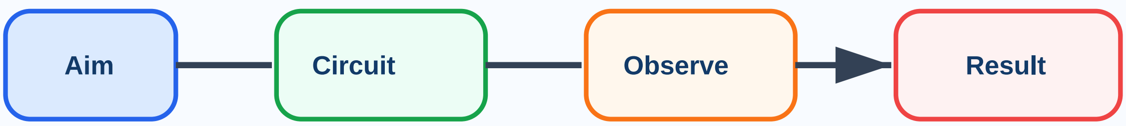
PWM modulator using 555 IC: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

PPM modulator using 555: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Input-output waveform comparison: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Duty cycle and pulse position observation: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Visual block / diagram idea



Pulse modulation circuits: lab record and practical workflow

Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

Practical example: For AM generation, set carrier and modulating signals, observe output envelope, measure V_{max} and V_{min}, and calculate modulation index $m = (V_{max} - V_{min})/(V_{max}+V_{min})$.

Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

Expected Questions

M3-Q1 Write the aim and apparatus required for an experiment based on Pulse modulation circuits.

M3-Q2 Explain the step-by-step procedure for Pulse modulation circuits.

M3-Q3 Draw or describe the required circuit/setup and observation table for Pulse modulation circuits.

M3-Q4 List important precautions and safety points while doing Pulse modulation circuits.

M3-Q5 Write viva questions with answers for Pulse modulation circuits.

M3-Q6 How will you write the final result and verify correctness in Pulse modulation circuits?

Module checklist

✓ Aim and apparatus are clear.

✓ Circuit/setup is neat and labelled.

✓ Procedure is in correct order.

✓ Observation table and result are complete.

✓ Precautions and viva points are prepared.

MODULE 4 • 15 HOURS

AM/FM stages and open-ended FM receiver

Build pre-emphasis, de-emphasis, frequency multiplier, mixer, audio amplifier and FM receiver setup.

Learning outcome

Build pre-emphasis, de-emphasis, frequency multiplier, mixer, audio amplifier and FM receiver setup.

Malayalam support: □□□□□ meaning □□□□□□□□□□□□□□, □□□□□□ English technical terms □□□□□□□□□□ answer □□□□□□.

Important terms

- Pre-emphasis circuit
- De-emphasis circuit
- Frequency multiplier using IC 565
- Mixer circuit using discrete components
- Audio power amplifier circuit
- FM receiver setup in 80 MHz-108 MHz range

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M4.01	Build pre-emphasis and de-emphasis circuits and plot characteristics	3	Applying
M4.02	Construct frequency multiplier using IC 565	3	Applying
M4.03	Build mixer using discrete components	3	Applying
M4.04	Build audio power amplifier circuit	3	Applying
Open-ended	Setup FM receiver in range 80 MHz - 108 MHz	3	Applying

Topic-wise explanation

Pre-emphasis circuit: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

De-emphasis circuit: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

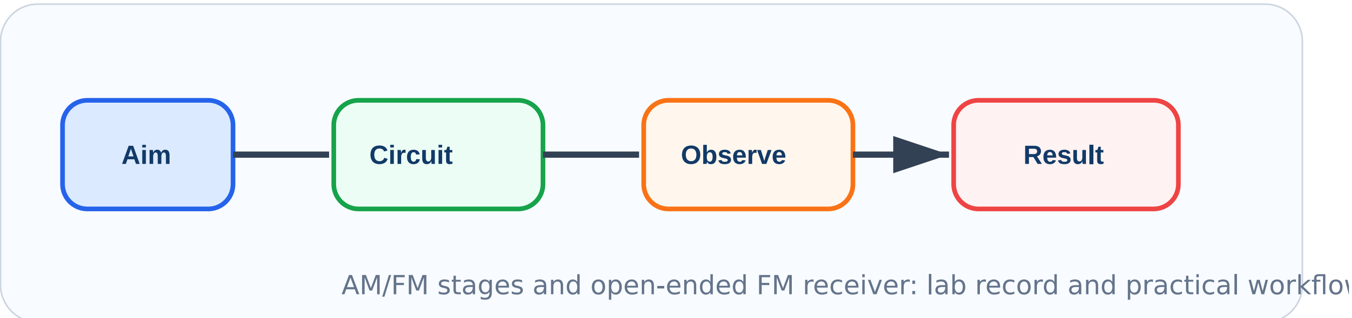
Frequency multiplier using IC 565: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Mixer circuit using discrete components: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Audio power amplifier circuit: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

FM receiver setup in 80 MHz-108 MHz range: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Visual block / diagram idea



Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

Practical example: For AM generation, set carrier and modulating signals, observe output envelope, measure V_{max} and V_{min}, and calculate modulation index $m = (V_{max}-V_{min})/(V_{max}+V_{min})$.

Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

Expected Questions

M4-Q1 Write the aim and apparatus required for an experiment based on AM/FM stages and open-ended FM receiver.

M4-Q2 Explain the step-by-step procedure for AM/FM stages and open-ended FM receiver.

M4-Q3 Draw or describe the required circuit/setup and observation table for AM/FM stages and open-ended FM receiver.

M4-Q4 List important precautions and safety points while doing AM/FM stages and open-ended FM receiver.

M4-Q5 Write viva questions with answers for AM/FM stages and open-ended FM receiver.

M4-Q6 How will you write the final result and verify correctness in AM/FM stages and open-ended FM receiver?

Module checklist

✓ Aim and apparatus are clear.

✓ Circuit/setup is neat and labelled.

✓ Procedure is in correct order.

✓ Observation table and result are complete.

✓ Precautions and viva points are prepared.

QUICK REVISION

One-page Revision Summary

Use this page in the last 24 hours before exam or lab evaluation.

Module-wise must-know points

1. **Module 1 - Communication components familiarization:** Know IC purpose, pins, ratings and typical communication applications.
2. **Module 2 - Analog modulation experiments:** Prepare AM/FM experiment aims, circuits, waveforms, modulation index and precautions.
3. **Module 3 - Pulse modulation circuits:** Know difference between PAM, PWM and PPM with waveform observations.
4. **Module 4 - AM/FM stages and open-ended FM receiver:** Revise circuit purpose, procedure, observations and open-ended FM receiver report.

Important definitions / keywords

AM generator · AM demodulator · FM generator · PLL · PAM · PWM · PPM · pre-emphasis · de-emphasis · mixer · audio power amplifier · FM receiver

Important formulas / key points

- Aim -> Apparatus -> Circuit -> Procedure -> Observation -> Result
- m=(Vmax-Vmin)/(Vmax+Vmin)
- Timer / pulse generator
- PLL IC
- 80 MHz - 108 MHz

Common exam mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

Last-minute checklist

✓ Memorise lab record format: aim, apparatus, theory, diagram, procedure, observation, result.

✓ Practise neat wiring/setup explanation.

✓ Prepare safety precautions and viva answers.

✓ Know how to calculate/read final result from observations.

Diagram / table list

- AM generator circuit
- AM diode demodulator circuit
- FM generator using 555
- FM generator/demodulator using 565
- PAM/PWM/PPM circuits
- Pre-emphasis/de-emphasis circuits
- Frequency multiplier block
- Mixer circuit
- Audio power amplifier circuit
- FM receiver setup block

MASTER ANSWER KEY

Master Answer Key - matched with Expected Questions

Every answer below matches the module expected questions exactly. Attempt first, then verify here.

M1-Q1 - Write the aim and apparatus required for an experiment based on Communication components familiarization.

Aim: Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M1-Q2 - Explain the step-by-step procedure for Communication components familiarization.

Aim: Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M1-Q3 - Draw or describe the required circuit/setup and observation table for Communication components familiarization.

Aim: Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M1-Q4 - List important precautions and safety points while doing Communication components familiarization.

Aim: Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M1-Q5 - Write viva questions with answers for Communication components familiarization.

Aim: Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M1-Q6 - How will you write the final result and verify correctness in Communication components familiarization?

Aim: Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q1 - Write the aim and apparatus required for an experiment based on Analog modulation experiments.

Aim: Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q2 - Explain the step-by-step procedure for Analog modulation experiments.

Aim: Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q3 - Draw or describe the required circuit/setup and observation table for Analog modulation experiments.

Aim: Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q4 - List important precautions and safety points while doing Analog modulation experiments.

Aim: Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q5 - Write viva questions with answers for Analog modulation experiments.

Aim: Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q6 - How will you write the final result and verify correctness in Analog modulation experiments?

Aim: Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q1 - Write the aim and apparatus required for an experiment based on Pulse modulation circuits.

Aim: Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q2 - Explain the step-by-step procedure for Pulse modulation circuits.

Aim: Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q3 - Draw or describe the required circuit/setup and observation table for Pulse modulation circuits.

Aim: Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q4 - List important precautions and safety points while doing Pulse modulation circuits.

Aim: Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q5 - Write viva questions with answers for Pulse modulation circuits.

Aim: Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q6 - How will you write the final result and verify correctness in Pulse modulation circuits?

Aim: Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q1 - Write the aim and apparatus required for an experiment based on AM/FM stages and open-ended FM receiver.

Aim: Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q2 - Explain the step-by-step procedure for AM/FM stages and open-ended FM receiver.

Aim: Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q3 - Draw or describe the required circuit/setup and observation table for AM/FM stages and open-ended FM receiver.

Aim: Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q4 - List important precautions and safety points while doing AM/FM stages and open-ended FM receiver.

Aim: Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q5 - Write viva questions with answers for AM/FM stages and open-ended FM receiver.

Aim: Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q6 - How will you write the final result and verify correctness in AM/FM stages and open-ended FM receiver?

Aim: Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.